Measuring and Mitigating Spatiotemporal Aliasing in Video Action Recognition: A Cross-Architecture Analysis at Scale

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Abstract. Temporal resolution in video action recognition is usually treated as a fixed hyperparameter rather than a sampling constraint. We present a large-scale study of empirical classifier-evidence loss under temporal and spatial subsampling across 8 architectures and 8 datasets (8,000 core evaluation configurations). Within this evaluated pool, aliasing sensitivity varies systematically across temporal aggregation designs, as certain sequence models and divided-attention Transformers suffer 3–5× less degradation under frame reduction compared to standard Convolutional Neural Networks (CNNs). We introduce the **Temporal Demand Score (TDS)** to quantify accuracy loss under temporal subsampling which has demonstrated that fine-grained sports demand as much temporal density as sign language. In a cross-resolution evaluation, CNNs degrade at novel resolutions without fine-tuning, whereas Transformers natively maintain accuracy. Furthermore, hardware profiling shows video state-space models are 4–8× slower than efficient Transformers. To optimize deployment, we demonstrate an **entropy-based adaptive frame routing** cascade that halves temporal budgets without accuracy loss. Finally, we provide a reproducibility package and an interactive visualization dashboard to enable aliasing-aware engineering design of human-observing AI systems.

Keywords: Temporal aliasing · Video recognition · Adaptive routing

1 Introduction

Applications of activity recognition include driving, healthcare, and surveillance [2, 13]. The field has evolved from 3D ConvNets and dual-pathway networks [4, 8, 29, 31, 35] to video Transformers and state-space models [1, 3, 7, 15, 18, 27, 33].

Despite this diversity, all models are evaluated using standardized sampling recipes (8–32 uniformly-sampled frames at fixed stride), treating temporal resolution as a static implementation detail. This assumption fails in real-world deployments where constraints on computational hardware, power usage, sensor capabilities, or communication bandwidth make dense sampling infeasible [17]. This leaves two fundamental questions unanswered: *when* does temporal subsampling become destructive, and *why* do different architectures respond differently?

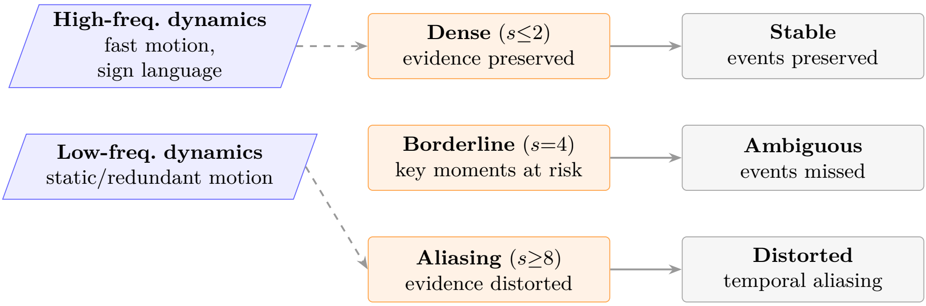

Fig. 1: Temporal aliasing as empirical evidence loss: sparse sampling can miss or distort the frames a recognizer needs, especially for temporally localized actions.

Sampling perspective. The Nyquist–Shannon theorem [23] motivates the intuition that undersampling can distort a temporal signal rather than merely compress it, a principle also relevant to perception [32] and deep learning [38]. We use this as a conceptual guide, but introduce an empirical Temporal Demand Score (TDS) metric that measures how much classifier evidence is lost under controlled subsampling, not literal spectral aliasing of the raw video signal.

Contributions. We conduct a systematic cross-architecture study of spatiotemporal aliasing:

1. **Temporal Demand Score (TDS)**: an architecture-averaged metric that measures a dataset's aliasing sensitivity. We demonstrate its utility by ranking 8 diverse datasets with leave-one-architecture-out analysis, architecture-family ablation (CNN-only, Transformer-only), and bootstrap confidence intervals checking ranking stability (Sec. 4.1; supplementary material).
2. **Cross-architecture aliasing characterization**: 8,000 core evaluation configurations show strong empirical architecture-level differences within our pool, with VideoMamba and TimeSformer losing 3–5$\times$ less accuracy under stride stress than CNNs and ViViT; this is an association, not a controlled temporal-module ablation (Sec. 4.2).
3. **Target-resolution fine-tuning**: Fine-tuning each model at the target spatial resolution closes the cross-resolution gap by up to 39pp for CNNs. In contrast, Transformers naturally maintain accuracy across different image sizes and require no retraining (Sec. 4.5)
4. **Measured inference latency**: GPU latency characterization across all 8 models $\times$ 5 target resolutions (bfloat16), enabling sampling-aware deployment budgeting per device class (Sec. 4.6).
5. **Practical illustration and reproducibility package**: as a low-cost downstream use of the temporal-sweep results above—rather than a new routing algorithm—a simple entropy-threshold cascade traces a validation Pareto frontier with 52% fewer frames on the Something-Something (SSv2) temporal dataset while matching dense-sampling accuracy. We also provide a supplementary package containing a deterministic dashboard, evaluation scripts, and result tables; public code will be released after review (Sec. 4.7).

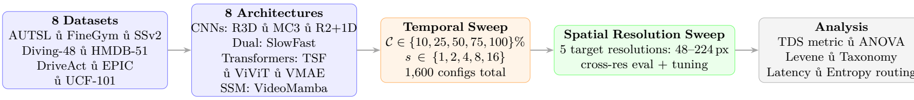

Fig. 2: Pipeline: the core factorial sweep uses 8 architectures × 8 datasets × 5 target resolutions (48–224px) × 5 coverages × 5 strides = **8,000 configurations**.

2 Related Work

2.1 Video Recognition Architectures

CNN-based architectures. 3D ConvNets learn spatiotemporal features end-to-end [4, 29], while Non-local Networks introduce global self-attention [35]. Factorized variants such as R(2+1)D and MC3 improve optimization and efficiency [30, 31]; SlowFast processes two frame-rate streams simultaneously [8].

Transformer and SSM architectures. Video Transformers extend patch-based recognition to time: TimeSformer uses divided space-time attention [3], ViViT uses factorized attention [1], and VideoMAE learns via masked patch reconstruction [27, 33]. VideoMamba [15] brings selective state-space scanning [6, 11] to video. Prior robustness work benchmarks spatial and photometric corruptions [21]; we instead isolate temporal and spatial sampling effects across this architectural spectrum.

2.2 Temporal Sampling and Adaptive Inference

Efficient inference methods sample or process fewer frames. TSN and TSM define sparse-sampling recipes [17, 34]; AdaFrame selects frames [37]; AdaFocus crops regions [36]; AR-Net adapts resolution [20]; and FrameExit uses confidence cascades [9]. Dataset-level temporal reasoning has been studied with shuffled frames [12] and motion-only cues [22]. These works do not measure the accuracy cliff induced by controlled stride and coverage sweeps; our TDS metric provides an architecture-averaged temporal-demand ranking.

2.3 Sampling Theory in Deep Learning

Sampling theory predicts that undersampling a signal below its highest temporal frequency creates aliasing [23]. Temporal aliasing is well documented in perception [32]; in deep learning, strided operators can violate shift-invariance unless anti-aliased [38]. Frequency-domain analysis has been explored for action recognition [28], but not as a large-scale cross-architecture stress test of modern video models.

3 Methodology

3.1 Datasets and Architectures

Table 1 summarizes the 8 datasets selected to span semantic domains, motion frequencies, and recording conditions. We evaluate 8 architectures from four

Table 1: TDS = mean accuracy drop (stride=1→16, cov=100%) per Eq. 1, averaged over all 8 architectures (n=8). Sorted by TDS. **Bold** = the two co-equal highest-demand domains (their relative order is leave-one-out sensitive; ranks 3–8 are stable under the supplementary robustness analysis).

Dataset	Classes	Videos	Domain	TDS	n
FineGym [24]	97	1,960	Fine-grained sport	**55.9pp**	8
AUTSL [25]	226	4,418	Sign language	**53.0pp**	8
SSv2 [10]	174	3,464	Causal/temporal	27.6pp	8
DriveAct [19]	33	448	In-vehicle	21.9pp	8
Diving-48 [16]	48	743	Fine-grained	19.2pp	8
HMDB-51 [14]	51	885	Sports	16.6pp	8
EPIC-Kitchens [5]	89	1,020	Egocentric	9.7pp	8
UCF-101 [26]	101	2,020	Appearance	4.9pp	8

families: (i) **3D CNNs**: R3D-18, MC3-18 [30], R(2+1)D-18 [31]; (ii) **dual-pathway**: SlowFast-R50 [8]; (iii) **Transformers**: TimeSformer [3], ViViT [1], VideoMAE [27]; (iv) **SSM**: VideoMamba [15]. All models are fine-tuned from Kinetics-400 checkpoints using AdamW with cosine decay. We use fixed validation clip lists across models (approximately 20 clips/class, or all available clips for smaller classes); additional split, crop, seed, and frame-normalization details are in the supplementary material.

3.2 Temporal Sampling Protocol

We decouple *temporal coverage* $\mathcal{C}$ (fraction of clip observed) from *temporal stride* s (sampling density), forming a 5×5 grid: $\mathcal{C} \in \{10, 25, 50, 75, 100\}\%$ and $s \in \{1, 2, 4, 8, 16\}$. The temporal sweep at native resolution yields 8×8×25=**1,600** model–dataset–configuration triples. Repeating this grid at the 5 target resolutions from 48–224px yields the **8,000** core configurations.

Coverage implementation. Coverage is applied from the *beginning* of each clip: the first $\lfloor T \cdot \mathcal{C}/100 \rfloor$ frames form the candidate pool, and s controls sampling density within that window. This simulates a sensor that observes only a leading fraction of the action and gives the sharpest measurement of how early the model has enough temporal context. Random window placement would confound temporal structure with missing-data effects; a 10% random-window check on Something-Something (SSv2) gives qualitatively similar coverage effects (supplementary material).

3.3 Spatial Resolution Protocol

Cross-resolution evaluation. The resolution grid is 48, 96, 112, 160, and 224 px. Native training resolution is 112px for CNNs and 224px for Transformers and VideoMamba.

Target-resolution fine-tuning. For each model–dataset–resolution triple at the 5 target resolutions (48–224 px), we fine-tune from the native-resolution checkpoint for 10 additional epochs. For Transformers and VideoMamba, positional embeddings are bicubically interpolated from the native grid to the target grid before fine-tuning. CNNs transfer directly, and SlowFast uses adaptive average pooling to support non-native resolutions.

3.4 Temporal Demand Score (TDS)

We define TDS as the architecture-averaged accuracy drop from stride=1 to stride=16:

$$\mathrm{TDS}(\mathcal{D}) = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} \big[\mathrm{Acc}(m, \mathcal{D}, s{=}1) - \mathrm{Acc}(m, \mathcal{D}, s{=}16)\big]^{+} \tag{1}$$

where $[x]^{+} = \max(x, 0)$, so a small negative drop (sparse sampling slightly outperforming dense sampling) contributes 0 rather than reducing the dataset score. We define feature collapse operationally as stride=1 Top-1 accuracy below 5%; no model–dataset pair in the native-resolution TDS computation met this criterion, so no entries are excluded and the ranking is unchanged. TDS is architecture-pool averaged, but its dataset ranking is stable under leave-one-architecture-out, family ablations, and bootstrap resampling over our 8-model pool (Sec. 4.1; supplementary material), suggesting it reflects dataset temporal structure rather than any single model.

3.5 Statistical Analysis

For each model-dataset pair, two-way ANOVA decomposes accuracy variance into coverage and stride components (effect size η^2). Levene's test detects per-class variance inflation under sparse sampling. Post-hoc Welch tests with Bonferroni correction identify aliasing cliffs.

3.6 Entropy-Based Routing

Given temporal-sweep results at two operating points—cheap: cov=100%/s=4 (4 effective frames); dense: cov=100%/s=1 (16 frames)—we route video v by threshold:

$$\hat{y}(v) = \begin{cases} \hat{y}_{\mathrm{cheap}} & \text{if } \max_c p(c \mid v, s{=}4) > \tau \\ \hat{y}_{\mathrm{dense}} & \text{otherwise} \end{cases}$$

We sweep $\tau \in [0.10, 0.95]$ on the validation split and report the Pareto-optimal point at avg frames ≤ 8. No retraining is required; only the existing temporal-sweep inference is reused. This reports an achievable validation frontier; deployment should select τ on a held-out calibration split.

4 Results

4.1 Temporal Demand Score Validates Architecture-Independent Ranking

Figure 3 (left) ranks datasets by TDS. **FineGym (55.9pp)** is the most important new finding in this dimension: fine-grained gymnastics phase transitions rival AUTSL sign language (53.0pp), showing that sport is not inherently low-TDS when the label space requires sub-second phase discrimination. AUTSL requires dense sampling to discriminate fine handshape transitions, but FineGym's phase boundaries are comparably demanding. UCF-101 (4.9pp) is nearly appearance-driven: object identity and static scene cues often suffice.

Crucially, ranks 3–8 are stable under leave-one-architecture-out: removing any single architecture leaves SSv2, DriveAct, Diving-48, HMDB-51, EPIC-Kitchens, and UCF-101 in the same order. Only the #1/#2 position swaps when VideoMamba is removed, so we report FineGym and AUTSL as co-equal high-TDS domains. This stability also holds for CNN-only, Transformer-only, and Transformer+SSM pools (Spearman $\rho \geq 0.976$ against the full pool), while bootstrap resampling gives a FineGym−AUTSL 95% CI of $[-4.6, +11.7]$pp, consistent with zero (supplementary material).

Spectral validation. Figure 3 (right) correlates optical-flow magnitude with per-class aliasing loss. AUTSL and DriveAct show the expected positive within-dataset trend: classes with more motion alias more. EPIC-Kitchens is near zero, suggesting that egocentric camera motion decouples raw flow from action-relevant demand. FineGym is inverted ($r = -0.475$, $p < 0.0001$): high-flow gymnastics classes often remain visually distinctive from almost any frame, while low-flow held positions or balance corrections depend on brief key moments. Thus raw flow magnitude is only a partial proxy for temporal demand, especially when class identity hinges on a discrete event rather than sustained motion (Table 2).

The inversion generalizes beyond FineGym. A spectral check on all 8 datasets across 5 resolutions pairs whole-frame optical-flow cutoff frequency with empirical stride-sensitivity at the same resolution (40 points; supplementary material). The relationship is significantly *negative* ($\rho = -0.63$, $p = 1.1\times10^{-5}$): UCF-101 and HMDB-51 have high flow-derived cutoffs but low stride-sensitivity, while FineGym and AUTSL have the highest stride-sensitivity at lower cutoff frequencies. Thus whole-frame flow does not validate a literal Nyquist account; it shows that classifier-relevant evidence is often localized and diluted by bulk motion. A supplementary sliding-window diagnostic is more consistent with this evidence-localization view: per-class correct-class confidence concentration is positively associated with per-class aliasing loss (pooled Spearman $\rho = +0.26$ over a 6-architecture ensemble).

4.2 Cross-Architecture Temporal Aliasing

Attention type correlates with aliasing. Table 3 shows stride=1 accuracy and stride=16 drop. The central observational finding is that aliasing aligns more

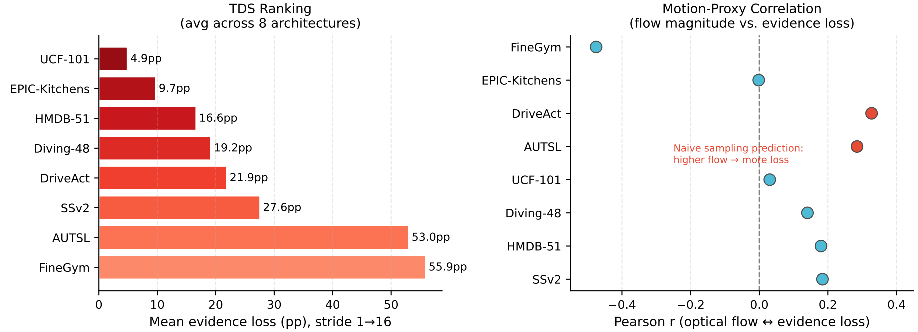

Fig. 3: Left: TDS ranking—mean accuracy drop from stride=1 to 16 averaged over all 8 architectures (n=8). FineGym (55.9pp) and AUTSL (53.0pp) are co-equal highest-demand domains (order between them is leave-one-out sensitive; see supplementary material); UCF-101 (4.9pp) is appearance-dominated. Positions 3–8 are exactly stable under leave-one-architecture-out. **Right**: Pearson correlation between optical-flow magnitude and per-class aliasing loss (8 datasets). AUTSL and DriveAct show the expected positive correlation; EPIC-Kitchens is near-zero (background camera motion confounds flow); FineGym shows the strongest correlation of any dataset but in the opposite direction ($r = -0.475$): discrete-event classes alias more than sustained-motion ones.

with temporal aggregation design than with broad architecture family in this pool, although many architectural and training factors covary. The robust group is TimeSformer (divided attention, 10.3pp mean drop) and VideoMamba (SSM, 11.4pp). Both appear to preserve global temporal context under aggressive subsampling: TimeSformer via full-clip cross-frame attention, and VideoMamba via recurrent state accumulation.

VideoMamba's robustness is markedly *bimodal*. It drops only 4.0pp on average over UCF-101, HMDB-51, EPIC-Kitchens, Diving-48, and DriveAct, which would make it the most robust model by a large margin if these were the only domains considered. But on the three highest-TDS domains, VideoMamba suffers substantially: FineGym (40.8pp), AUTSL (16.0pp), and SSv2 (14.1pp). FineGym most clearly inverts the SSM-vs-Transformer ranking: TimeSformer drops 36.1pp vs. VideoMamba's 40.8pp, a 4.7pp gap favoring divided attention. The same direction holds on SSv2, while AUTSL is tied. We hypothesize that VideoMamba's running state smooths over high-frequency phase transitions, whereas TimeSformer can selectively attend to the most discriminative frames.

The SSv2 result extends beyond temporal frequency: SSv2 tests causal ordering (e.g., move object left then right), not just temporal density. SSM recurrence over skipped frames integrates corrupted order evidence, while appearance-driven UCF-101 remains stable. The fragile group comprises SlowFast (42.1pp), VideoMAE (32.8pp), ViViT (30.0pp), R2+1D (30.0pp), R3D-18 (29.7pp), and MC3-18 (22.6pp). ViViT is the key anomaly: a Transformer whose factorized attention remains vulnerable to frame dropout.

Table 2: Spectral validation: Pearson r between optical-flow magnitude and per-class aliasing loss. Flow alone partially explains temporal demand; camera-motion-heavy datasets (EPIC, UCF) show weak correlation.

Dataset	Classes	Pearson r	Significant	Interpretation
FineGym	97	-0.475	✓	Discrete-event classes alias more
DriveAct	33	0.327	marginal	Flow $\sim$ arm motion
AUTSL	226	0.285	✓	Flow $\sim$ hand speed
SSv2	174	0.184	✓	Causal/object motion
HMDB-51	51	0.180	×	Mixed motion types
Diving-48	47	0.140	×	Body rotation dominated
UCF-101	101	0.031	×	Appearance dominates
EPIC-Kitchens	89	-0.002	×	Camera motion confounds

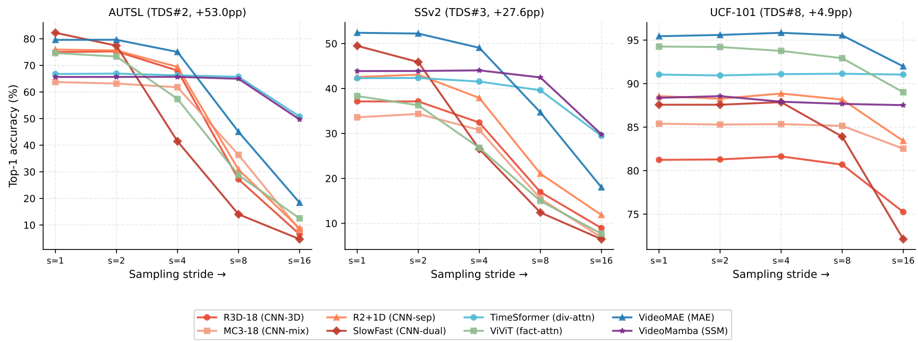

Fig. 4: Accuracy vs. stride (cov=100%) for three representative datasets. **AUTSL** (TDS=53.0pp): TimeSformer and VideoMamba remain most stable; all other models cliff above stride=4. **SSv2** (TDS=27.6pp): clear two-tier separation between robust (TSF, VMamba) and fragile models. **UCF-101** (TDS=4.9pp): minimal sensitivity across all architectures. These three datasets span the full TDS range.

Architecture-specific failure modes. SlowFast cliffs at stride 2→4 because its fast and slow pathways are simultaneously starved; on FineGym it has the best base accuracy (78.2%) but the largest drop (71.5pp). VideoMAE is spatially strong but temporally fragile: masked patch reconstruction does not train robustness to missing frames. VideoMamba's failures concentrate on high-frequency phase changes (FineGym), handshape transitions (AUTSL), and ordering-sensitive interactions (SSv2).

Cross-architecture sensitivity heatmap. Figure 5 shows the stride-induced accuracy drop (stride=1 minus stride=16, in percentage points) at cov=100%, with models sorted by mean drop. This is an *accuracy-drop* heatmap, not a Top-1 accuracy heatmap; larger values indicate stronger aliasing loss. Full per-model, per-dataset Top-1 accuracy heatmaps across coverage and stride are in the supplementary material.

Table 3: Full temporal aliasing table: stride=1 accuracy (%) / drop at stride=16 (pp), cov=100%. Rows sorted by average drop across all 8 datasets.

Model (family)	UCF-101	SSv2	HMDB-51	Diving-48	FineGym	AUTSL	DriveAct	EPIC-Kit.	Avg drop
TimeSformer (div. attn)	91.0/0.0	42.3/12.8	79.9/4.0	38.2/2.0	66.4/36.1	66.8/16.0	67.6/8.5	32.3/2.9	**10.3pp**
VideoMamba (SSM)	88.4/+0.8	43.9/14.1	69.8/3.6	36.5/4.9	72.7/40.8	65.6/16.0	57.6/9.6	28.3/1.2	**11.4pp**
MC3-18 (CNN-mix)	85.4/2.9	33.6/26.7	78.6/9.7	31.6/12.4	64.2/51.1	63.7/55.5	69.0/13.8	36.2/8.6	22.6pp
R3D-18 (CNN-3D)	81.2/6.0	37.1/28.2	80.3/21.1	28.8/16.0	71.5/61.4	75.0/68.5	68.3/23.9	37.2/12.7	29.7pp
R2+1D (CNN-sep.)	88.6/5.1	42.6/30.7	73.1/17.6	35.3/18.8	76.8/64.2	75.9/67.2	62.5/24.8	35.5/11.7	30.0pp
ViViT (fact. attn)	94.3/5.2	38.3/30.6	80.2/19.4	53.0/36.3	69.9/55.2	74.6/62.1	67.4/20.3	32.9/11.1	30.0pp
VideoMAE (MAE)	95.4/3.5	52.4/34.4	84.0/20.2	48.6/23.4	78.0/67.8	79.6/61.2	73.9/40.8	37.7/10.8	32.8pp
SlowFast (dual)	87.6/15.4	49.5/43.0	79.3/37.4	50.5/39.7	78.2/71.5	82.3/77.6	72.5/33.7	39.4/18.5	**42.1pp**

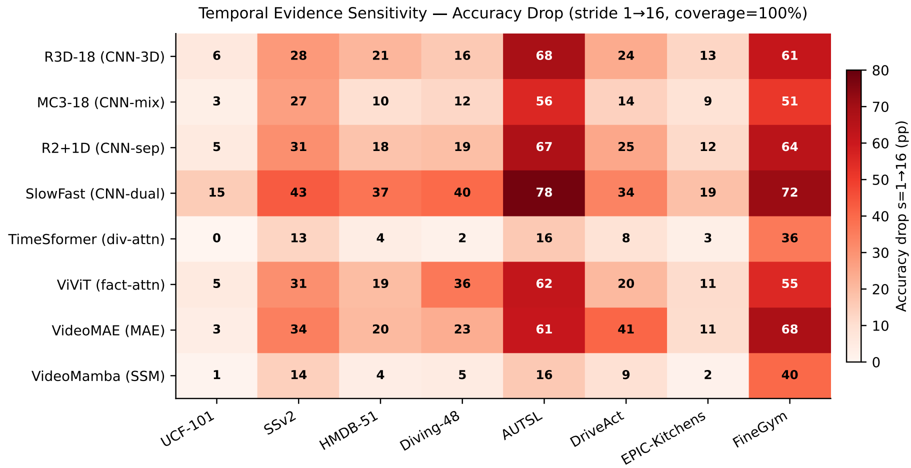

Fig. 5: Cross-architecture aliasing-loss heatmap: accuracy drop (percentage points) from stride=1 to stride=16 at cov=100%. Models are sorted left-to-right by increasing mean drop. Low values mean temporal robustness; high values mean strong stride-induced aliasing loss.

4.3 Statistical Validation (ANOVA and Levene's Test)

Effect sizes. Table 4 reports two-way ANOVA η^2 values (mean±std over 8 datasets). Coverage is the dominant factor across all models ($\eta^2_{\text{cov}} = 0.53$–$0.90$). Stride effect size separates the empirical groups: VideoMamba and TimeSformer have $\eta^2_{\text{stride}} \approx 0.08$, while SlowFast reaches 0.35—a 4.4× difference. Full per-dataset η^2 tables are provided in the supplementary material.

Variance inflation. Levene's test shows significant inter-class variance inflation in 67% of model-dataset pairs ($p < 0.05$, Bonferroni-corrected across all 64 model×dataset comparisons). The worst cases are VideoMAE/HMDB-51 (2.0×) and R3D-18/HMDB-51 (1.85×), while TimeSformer and VideoMamba remain stable ($\approx 1.1\times$). Thus sparse sampling can create class-specific failures that mean accuracy hides.

Table 4: ANOVA effect sizes (η^2, mean±std over 8 datasets). All stride effects significant (p<0.05). Coverage dominates; stride separates robust from fragile.

Model	η^2_{stride}	η^2_{cov}	Interpretation
VideoMamba	0.09 ± 0.14	0.88 ± 0.07	Robust
TimeSformer	0.11 ± 0.13	0.88 ± 0.06	Robust
MC3-18	0.20 ± 0.07	0.73 ± 0.10	Moderate
R2+1D	0.23 ± 0.08	0.69 ± 0.09	Fragile
R3D-18	0.24 ± 0.06	0.68 ± 0.07	Fragile
VideoMAE	0.20 ± 0.11	0.73 ± 0.14	Moderate
ViViT	0.26 ± 0.05	0.65 ± 0.09	Fragile
SlowFast	0.35 ± 0.07	0.53 ± 0.09	Most fragile

Table 5: Taxonomy tier boundaries (accuracy drop pp): classes below Low threshold alias minimally; above High threshold alias severely. Computed at cov=100%, stride=1→16.

Dataset	Low ≤	High >	# High	# Low
AUTSL	66.9pp	73.9pp	75	74
Diving-48	25.0pp	47.7pp	16	16
SSv2	59.7pp	76.9pp	58	57
HMDB-51	8.4pp	28.5pp	17	17
EPIC-Kitchens	0.0pp	19.4pp	30	11
DriveAct	14.8pp	51.0pp	11	11
UCF-101	1.4pp	7.5pp	35	31

4.4 Action Sensitivity Taxonomy

We classify each action class into three tiers by per-class aliasing loss (accuracy drop, stride=1→16). Tier boundaries use the 33rd/67th percentiles.

High-sensitivity actions are temporally critical: in AUTSL, even the Low tier loses 38pp, confirming that the entire dataset requires dense sampling. In contrast, the UCF-101 Low tier loses only 0.3pp: static, appearance-driven actions are virtually unaffected. Table 5 reports per-dataset tier boundaries. AUTSL's Low threshold starts at 66.9pp, above the High threshold of any other dataset, confirming that sign language forms a distinct temporal-frequency regime.

Clip duration effect. Counter-intuitively, shorter clips alias more severely (Pearson $r = -0.3$ to -0.8 across models): short clips have less temporal redundancy, so each missed frame removes a larger fraction of available evidence. On UCF-101, clips >6s lose only 1.5pp at stride=16 vs. 16.2pp for clips <3s. This suggests that short clips should receive dense sampling irrespective of model confidence (details in the supplementary material).

4.5 Spatial Resolution and Target-Resolution Fine-Tuning

Cross-resolution robustness. Figure 6 and Table 6 show accuracy vs. input resolution for all 8 models on SSv2, evaluated *without retraining*. A related empirical split appears in the spatial domain. CNNs (native 112px) degrade sharply at higher resolutions: R2+1D drops from 42.6% at 112px to 20.8% at 224px

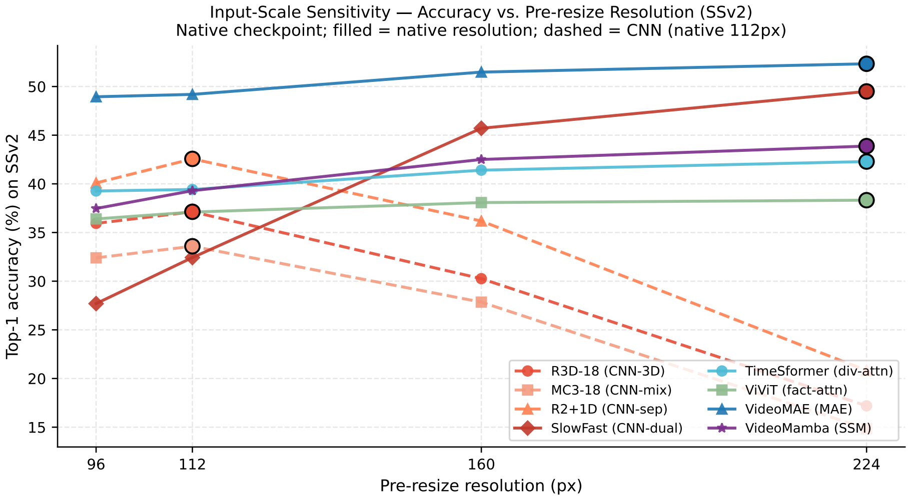

Fig. 6: Accuracy vs. input resolution on SSv2 (no retraining). CNNs (dashed, native=112px) degrade at off-native resolutions; Transformers and VideoMamba (solid, native=224px) stay within ±7pp across 96–224px. Filled markers denote each model's native training resolution.

Table 6: Spatial robustness on SSv2 (no retraining). Bold = native resolution. $\Delta_{\max}$ = largest drop from native.

Model	Native	96px	112px	160px	224px	$\Delta_{\max}$
R3D-18	112px	35.9	**37.1**	30.3	17.2	−19.9pp
MC3-18	112px	32.4	**33.6**	27.8	14.8	−18.8pp
R2+1D	112px	40.1	**42.6**	36.2	20.8	−21.8pp
SlowFast	224px	27.7	32.4	45.7	**49.5**	−21.8pp
TimeSformer	224px	39.3	39.4	41.4	**42.3**	−3.0pp
ViViT	224px	36.4	37.1	38.1	**38.3**	−1.9pp
VideoMAE	224px	48.9	49.2	51.5	**52.3**	−3.4pp
VideoMamba	224px	37.5	39.3	42.5	**43.9**	−6.4pp

(−21.8pp). Transformers and VideoMamba (native 224px) stay within ±7pp across 96–224px.

This spatial robustness pattern is similar across the 8 datasets (supplementary material). The largest off-native CNN drop in the 48–224px grid is 62.8pp on DriveAct, whereas Transformers remain within ±8pp on 7 of 8 datasets; EPIC-Kitchens is the main exception.

Target-resolution fine-tuning recovers CNN performance. Table 7 shows family-level averages after target-resolution tuning. CNNs recover sharply: at 224px, cross-resolution accuracy is 30.0%, but fine-tuning reaches 69.2% (+39.2pp). Transformers improve modestly because they are already robust across resolutions.

Table 7: Target-resolution fine-tuning: family-level mean accuracy (%) averaged over all 8 datasets, after fine-tuning from the native-resolution checkpoint for 10 epochs at each target spatial resolution. CNNs benefit by up to 39.2pp at off-native resolutions. Transformers already achieve near-native accuracy without retraining.

Family	48px	96px	112px	160px	224px
CNN (3 models)	49.3	64.9	67.7	68.5	69.2
Transformer (3)	36.6	56.3	60.3	66.9	71.4
SSM (VideoMamba)	31.5	41.2	45.1	54.5	59.7
Dual-CNN (SlowFast)	36.2	73.3	71.9	71.6	67.5

Table 8: Inference latency (ms/clip, bfloat16, batch=1, RTX PRO 6000 Blackwell). VideoMamba is 4–8× slower than VideoMAE despite being an SSM. Scale to other devices: RTX 4090 ×1.5, Jetson AGX Orin ×10, Jetson Nano ×50.

Model	48px	96px	112px	160px	224px
R3D-18	0.61	0.89	1.20	1.76	2.78
MC3-18	0.61	0.85	1.18	1.84	3.15
R2+1D	1.39	2.21	2.54	4.10	5.77
SlowFast	3.09	3.11	3.11	3.13	4.10
TimeSformer	3.25	3.24	3.20	3.20	4.30
ViViT	1.55	1.77	1.99	3.19	6.35
VideoMAE	1.57	1.79	1.78	1.92	2.82
VideoMamba	8.06	8.15	8.18	10.03	15.44

4.6 Inference Latency and Deployment Budgeting

Table 8 reports measured bfloat16 latency at batch 1. VideoMamba is slowest (8–15ms), 4–8× slower than VideoMAE (1.6–2.8ms) despite its SSM design. VideoMAE gives the best Transformer accuracy per ms, while CNN latency scales predictably with resolution.

4.7 Entropy-Based Adaptive Routing

Table 9 compares entropy routing at average frame budget ≤ 8 (full curves in the supplementary material). Thresholds are swept on the validation split to trace the achievable Pareto frontier; we treat the router as a deployment illustration rather than a separately trained method. AdaFocus [36] and AR-Net [20] use ResNet-50 and are informational. The controlled comparison is FrameExit [9], also on TimeSformer: entropy routing reaches 42.5% vs. 38.4% (+4.2pp) with fewer frames (7.7 vs. 9.9).

Across validation model-dataset pairs, 77% of videos route to cheap 4-frame inference on average. VideoMAE gives the highest absolute routed accuracy, while TimeSformer achieves the highest cheap-routing rates on temporal-critical datasets.

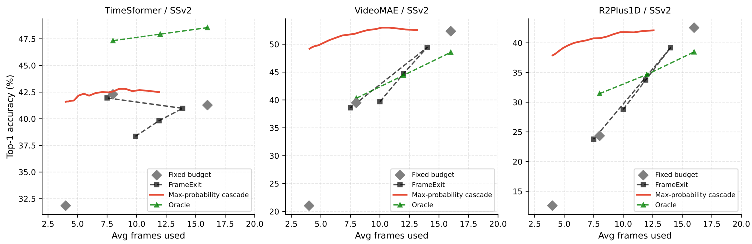

Fig. 7: Entropy routing on SSv2: accuracy vs. average frames as τ sweeps between 4-frame cheap and 16-frame dense inference.

Table 9: Routing comparison on SSv2 at avg $\leq$8 frames. The fair, same-backbone comparison is Entropy (ours) vs. FrameExit, both on TimeSformer. AdaFocus/AR-Net use a different backbone (ResNet-50) and are reported only as context for absolute numbers reported in their respective papers, not as a controlled comparison.

Method	Avg frames	SSv2 Top-1	Backbone
Informational only: different backbone			
AdaFocus [36]	8.0	54.7%	ResNet-50
AR-Net [20]	8.0	48.6%	ResNet-50
Controlled comparison: same TimeSformer backbone			
Oracle upper bound	7.7	47.3%	TimeSformer
Fixed 16f (dense)	16.0	42.3%	TimeSformer
Fixed 8f	8.0	42.3%	TimeSformer
Entropy (ours)	**7.7**	**42.5%**	TimeSformer
Fixed 4f (cheap)	4.0	41.5%	TimeSformer
FrameExit [9]	9.9	38.4%	TimeSformer

5 Discussion

Architecture and sampling. Within our evaluated pool, the robust temporal group combines global divided attention (TimeSformer) or recurrent state integration (VideoMamba), while the fragile group includes CNNs and factorized-attention Transformers. VideoMamba remains bimodal: it is nearly flat on low-TDS datasets but still aliases on FineGym, AUTSL, and SSv2, where brief phase changes, handshape transitions, or causal order matter. Spatial robustness follows a related but distinct pattern: patch-token Transformers are resolution-stable, whereas CNNs require target-resolution fine-tuning; ViViT's spatial robustness despite temporal fragility separates tokenization effects from temporal aggregation effects.

Spatial robustness. The spatial results suggest that temporal and spatial robustness share some architectural ingredients but are not identical. CNNs' strided filters are tied to the spatial scale seen during training, which explains the sharp off-resolution degradation and the large recovery from target-resolution fine-tuning. Transformers remain stable across most resolutions because patch tokenization and global aggregation tolerate scale shifts, yet ViViT shows that this spatial invariance does not guarantee temporal robustness. VideoMamba again sits between these regimes: stable on some datasets, but less reliable on cluttered egocentric or in-vehicle scenes.

Deployment implications. Resolution, latency, and sampling density should be searched jointly rather than treated as independent knobs. High-TDS domains require stride$\leq$2, moderate-TDS domains tolerate stride$\leq$4, and low-TDS domains tolerate stride up to 16. Target-resolution tuning makes CNN resolution a cost dial; TimeSformer routing at 7.7 frames matches dense R2+1D on SSv2. These trade-offs are deployment-relevant because a model that is slightly less accurate at dense sampling can become preferable once resolution, stride, and hardware latency are optimized together.

This also changes how one should read fixed-recipe video benchmarks. A single native-resolution, fixed-stride number can hide the fact that two models exchange rank once the temporal budget or input resolution changes. The dashboard and tables therefore expose operating points rather than only peak accuracy, making the sampling constraint explicit.

Limitations. Our architecture analysis is observational: temporal aggregation design covaries with tokenization, pretraining objective, capacity, and optimization recipe, so we report empirical associations rather than a controlled causal mechanism. A stronger test would swap temporal modules inside a shared backbone while holding training and capacity fixed. The Nyquist framing is conceptual: whole-frame flow is inversely related to stride-sensitivity, while temporal concentration of correct-class confidence is positively associated with per-class aliasing (pooled Spearman $\rho = +0.26$; supplementary material). Routing thresholds trace validation Pareto frontiers and should be calibrated on a held-out split before deployment.

6 Conclusion

We present a large-scale cross-architecture study of spatiotemporal aliasing in video action recognition. Across 8 architectures, 8 datasets, and 8,000 configurations, aliasing shows a strong empirical association with temporal aggregation design, with ViViT showing that Transformers are not automatically temporally robust. The TDS metric identifies FineGym and AUTSL as co-equal high-demand domains, establishing fine-grained sport as a high-aliasing regime. VideoMamba is robust but bimodal and slow on GPU; target-resolution fine-tuning closes the CNN resolution gap by up to 39pp; and a validation-set entropy cascade illustrates a 52% lower-frame operating point on SSv2. These results are useful beyond ranking models: they indicate when computation should be spent on denser temporal evidence, when frames can be removed with little cost, and when poor accuracy is caused by a sampling mismatch rather than by the recognizer alone. Reporting accuracy over stride, coverage, and resolution, therefore, gives practitioners a way to check whether an activity model has actually seen enough time to justify its prediction.

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